Every prescribed least period in a complex rational delay recurrence

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Abstract

For every integer $p \geq 2$, we construct an everywhere-defined solution of least period p of $z_{n+1} = (\alpha + \beta z_{n-k})/(\gamma - z_n)$, with $k = p - 1$ and all three parameters nonzero and nonreal. The construction proves the higher-period existence conjecture in MathDB 333488.

Task 1: a full construction for every period

Conjecture 3.1 on page 14 of Hassan's manuscript [1] asks for arbitrarily high periods of this family of delay recurrences. Pages 1–2 allow arbitrary complex parameters and initial conditions, and the examples immediately preceding the conjecture vary the delay and parameters. No fixed-delay or fixed-parameter restriction is stated. We prove the precise stronger existence statement that every prescribed integer $p \geq 2$ occurs as a least period, with explicit parameters, delay, and initial history. This is an existence result in the family; it makes no claim that one fixed recurrence realizes all periods.

Theorem 1. *Fix an integer $p \geq 2$ and put*

$$\theta = \frac{\pi}{p}, \qquad t = 2\cos\theta, \qquad s = 1 + i, \qquad k = p - 1,$$

$$\alpha = s^2 = 2i, \qquad \beta = s = 1 + i, \qquad \gamma = s(1 + t).$$

Let $T(w) = 1/(t - w)$, set $w_0 = 2i$, and define $w_j = T^j(w_0)$ for $j \geq 0$. Then w_j is periodic with least period p. Extend it periodically to all integer indices and set $z_j = sw_j$. The initial history $z_{-k}, \ldots, z_0$ produces an everywhere-defined solution of

$$z_{n+1} = \frac{\alpha + \beta z_{n-k}}{\gamma - z_n} \qquad (n \geq 0)$$

of least period p. All three parameters are nonzero and nonreal.

Proof. First, every w_j lies in the upper half-plane. Indeed, t is real and

$$\operatorname{Im} T(w) = \frac{\operatorname{Im} w}{|t - w|^2} > 0 \qquad (\operatorname{Im} w > 0).$$

Thus every iterate is finite and the denominator $t - w_j$ never vanishes.

Set $u = e^{i\theta}$ and $v = e^{-i\theta}$. These are distinct, satisfy $u + v = t$ and $uv = 1$, and are the fixed points of T. Direct subtraction gives

$$T(w) - u = \frac{u(w - u)}{t - w}, \qquad T(w) - v = \frac{v(w - v)}{t - w}.$$

Consequently the injective fractional linear map $R(w) = (w-u)/(w-v)$ satisfies

$$R(T(w)) = e^{2i\theta} R(w).$$

This identity is defined throughout the orbit because v lies in the lower half-plane. Also $R(w_0) \neq 0$, since $|w_0| = 2$ and $|u| = 1$. It follows by induction that

$$R(w_j) = e^{2\pi i j/p} R(w_0).$$

The multiplier is a primitive pth root of unity, so injectivity of R gives $w_j = w_0$ if and only if p divides j. In particular, $w_p = w_0$, and iteration gives $w_{j+p} = w_j$ for all $j \geq 0$. This proves periodicity and minimality, not just an upper bound on the period. The periodic extension also satisfies $w_{j+1} = T(w_j)$ for all integers j.

Now $k = p - 1$ implies $w_{n-k} = w_{n+1}$. The defining equation of T is $w_{n+1}(t - w_n) = 1$, and hence

$$1 + w_{n-k} = 1 + w_{n+1} = (1 + t - w_n) w_{n+1}.$$

Because w_n is nonreal, the factor $1+t-w_n$ is nonzero. Substituting the stated parameters therefore gives

$$\frac{\alpha + \beta z_{n-k}}{\gamma - z_n} = \frac{s^2 + s^2 w_{n-k}}{s(1 + t - w_n)} = s w_{n+1} = z_{n+1}.$$

No pole is introduced, since $s \neq 0$. Multiplication by s preserves least period. Finally $0 \leq t < 2$, so $1 + t > 0$ and the imaginary parts of α, β, γ are respectively $2, 1, 1 + t$, all positive. □

The use of p as the least period agrees with the source's "prime period" definition. In particular the result supplies arbitrarily large periods without counting multiples of a shorter period. The construction does not assert attraction or chaos.

Four proposed follow-up tasks

Task 2. Classify the parameters for which the ansatz $z_{n-k} = z_{n+1}$ reduces the recurrence to a finite-order fractional linear transformation, retaining all pole exclusions.

Task 3. For each prescribed period, determine the smallest positive delay that admits a solution of that least period.

Task 4. Determine the stability of these periodic solutions under perturbations of the full initial history of length $k + 1$.

Task 5. Decide whether a single fixed parameter triple and fixed delay can admit periodic solutions of unbounded least period.

References

[1] Sk. Sarif Hassan, *Dynamics of the complex rational delay recursive sequence* $z_{n+1} = (\alpha + \beta z_{n-k})/(\gamma - z_n)$, arXiv:1606.08887v1 (2016), pp. 1–2 and Conjecture 3.1, p. 14.